

# Numerical Method

## Bisection Method

### Algorithm of Bisection Method:

- 1) Define a function  $f(x)$ .
- 2) Find two points  $a$  and  $b$  so that  $f(a) < 0$  and  $f(b) > 0$ .
- 3) Define a new point,  $c = (a + b) / 2$ .
- 4) If  $f(c) < 0$ , then set  $a = c$ , otherwise set  $b = c$ .
- 5) Continuing the process till the desired place of decimals is not obtained.

### Question - 01:

$x^2 - 3 = 0$  for  $x \in [1, 2]$  by using Bisection Method.

$\Rightarrow$  Given that,  $x^2 - 3 = 0$ , let  $f(x) = x^2 - 3$

Now, find the value of  $f(x)$  between  $a = 1$  and  $b = 2$ .

$$f(1) = 1^2 - 3 = 1 - 3 = -2 < 0$$

$$f(2) = 2^2 - 3 = 4 - 3 = 1 > 0$$

The given function is continuous and the root lies in the interval  $[1, 2]$

The iterations for the given function are:  $\epsilon = 10^{-3}$

| Iteration | a          | b         | $c = (a+b)/2$ | $f(c)$      |
|-----------|------------|-----------|---------------|-------------|
| 1         | 1.0        | 2.0       | 1.5           | -0.7500     |
| 2         | 1.5        | 2.0       | 1.75          | 0.0625      |
| 3         | 1.5        | 1.75      | 1.625         | -0.359375   |
| 4         | 1.625      | 1.75      | 1.6875        | -0.15234    |
| 5         | 1.6875     | 1.75      | 1.71875       | -0.0458     |
| 6         | 1.71875    | 1.75      | 1.734375      | 0.008057    |
| 7         | 1.71875    | 1.734375  | 1.7265625     | -0.01898    |
| 8         | 1.7265625  | 1.734375  | 1.73047       | -0.00547    |
| 9         | 1.73047    | 1.734375  | 1.7324225     | 0.0012877   |
| 10        | 1.73047    | 1.7324225 | 1.73144625    | -0.002938   |
| 11        | 1.73144625 | 1.7324225 | 1.73193437    | -0.00040332 |

Since,  $|x_{11} - x_{10}| = 4.8812 \times 10^{-4} < 10^{-3}$ . Hence, 1.73193437 is the approximated solution.

### Question - 02:

Given that,  $e^{-x}(3.2 \sin x - 0.5 \cos x) = 0$  for  $x \in [0, 0.5]$ .

by using let,  $f(x) = e^{-x}(3.2 \sin x - 0.5 \cos x)$

$$\text{for } f(0) = e^0(3.2 \sin 0 - 0.5 \cos 0) = -0.5 < 0$$

$$f(0.5) = e^{-0.5} \{3.2 \sin(0.5) - 0.5 \cos(0.5)\} = 0.664 > 0$$

The given function is continuous and the root lies in the interval  $[0, 0.5]$

The iteration for the given function are:  $\epsilon = 10^{-1}$

| Iteration | a           | b           | c = (a+b)/2 | f(c)        |
|-----------|-------------|-------------|-------------|-------------|
| 1         | 0           | 0.5         | 0.25        | 0.280596    |
| 2         | 0           | 0.25        | 0.125       | -0.08572    |
| 3         | 0.125       | 0.25        | 0.1875      | 0.08725     |
| 4         | 0.125       | 0.1875      | 0.15625     | 0.003471    |
| 5         | 0.125       | 0.15625     | 0.140625    | -0.04043    |
| 6         | 0.140625    | 0.15625     | 0.1484375   | -0.01831    |
| 7         | 0.1484375   | 0.15625     | 0.15234375  | -0.007378   |
| 8         | 0.15234375  | 0.15625     | 0.154296875 | -0.001912   |
| 9         | 0.154296875 | 0.15625     | 0.155273437 | 0.0007672   |
| 10        | 0.154296875 | 0.155273437 | 0.154785156 | -0.000587   |
| 11        | 0.154785156 | 0.155273437 | 0.155029296 | -0.00009029 |
| 12        | 0.155029296 | 0.155273437 | 0.155151366 | 0.00012207  |
| 13        | 0.155029296 | 0.155151366 | 0.155090331 | 0.0002597   |

So, 0.15509 is the approximated solution.

## Exercise

### Q - 01:

$$x^3 + 4x^2 - 10 = 0; x \in [1, 2]; \epsilon = 10^{-4}$$

Soln: Given that,  $x^3 + 4x^2 - 10 = 0$ , let  $f(x) = x^3 + 4x^2 - 10$

$$f(1) = (1)^3 + 4 \cdot (1)^2 - 10 = -5 < 0$$

$$f(2) = (2)^3 + 4 \cdot (2)^2 - 10 = 14 > 0$$

This function is continuous, and the root lies in the interval  $[1, 2]$ .

The iteration for the given functions are:  $\epsilon = 10^{-4}$

| Iteration | a           | b           | c = (a+b)/2 | f(c)       |
|-----------|-------------|-------------|-------------|------------|
| 1         | 1.00        | 2.00        | 1.5         | 2.375      |
| 2         | 1.00        | 1.5         | 1.25        | -1.7968    |
| 3         | 1.25        | 1.5         | 1.375       | 0.1621     |
| 4         | 1.25        | 1.375       | 1.3125      | -0.8483    |
| 5         | 1.3125      | 1.375       | 1.34375     | -0.3509    |
| 6         | 1.34375     | 1.375       | 1.359375    | -0.0964    |
| 7         | 1.359375    | 1.375       | 1.3671875   | 0.03235    |
| 8         | 1.359375    | 1.3671875   | 1.36328125  | -0.03214   |
| 9         | 1.36328125  | 1.3671875   | 1.365234375 | 0.00007202 |
| 10        | 1.36328125  | 1.365234375 | 1.364257813 | -0.016096  |
| 11        | 1.364257813 | 1.365234375 | 1.364746094 | -0.007989  |
| 12        | 1.364746094 | 1.365234375 | 1.364990235 | -0.003959  |
| 13        | 1.364990235 | 1.365234375 | 1.365112305 | -0.00194   |
| 14        | 1.365112305 | 1.365234375 | 1.36517334  | -          |

$$\therefore |x_{14} - x_{13}| = |1.36517334 - 1.365112305| = 6.1 \times 10^{-5} < 10^{-4}$$

So, 1.36517334 is the approximate solution.

**Ques - 02:**

$$x^3 - x - 1 = 0; \varepsilon = 10^{-4}$$

Soln: Given that,  $x^3 - x - 1 = 0$ , let  $f(x) = x^3 - x - 1$

$$f(0) = (0)^3 - 0 - 1 = -1 < 0$$

$$f(1) = (1)^3 - 1 - 1 = -1 < 0$$

$$f(2) = (2)^3 - 2 - 1 = 8 - 2 - 1 = 5 > 0$$

$$\text{Since, } f(1) \times f(2) = -1 \times 5 = -5 < 0$$

The function is continuous, and the root lies in interval  $[1, 2]$ .

The iteration of the functions are:  $\varepsilon = 10^{-4}$

| Iteration | a           | b           | c = (a+b)/2 | f(c)      |
|-----------|-------------|-------------|-------------|-----------|
| 1         | 1.00        | 2.00        | 1.5         | 0.875     |
| 2         | 1.00        | 1.5         | 1.25        | -0.296875 |
| 3         | 1.25        | 1.5         | 1.375       | 0.2246    |
| 4         | 1.25        | 1.375       | 1.3125      | -0.0515   |
| 5         | 1.3125      | 1.375       | 1.34375     | 0.08261   |
| 6         | 1.34375     | 1.375       | 1.328125    | 0.01457   |
| 7         | 1.3125      | 1.34375     | 1.320325    | -0.01865  |
| 8         | 1.320325    | 1.32815     | 1.3242375   | -0.002098 |
| 9         | 1.3242375   | 1.32815     | 1.32619375  | 0.006302  |
| 10        | 1.3242375   | 1.32619375  | 1.325215625 | 0.002123  |
| 11        | 1.3242375   | 1.325215625 | 1.324726563 | 0.0000257 |
| 12        | 1.3242375   | 1.324726563 | 1.324982032 | -0.001005 |
| 13        | 1.324482032 | 1.324726563 | 1.324604238 | -0.000184 |
| 14        | 1.324604238 | 1.324726563 | 1.324665431 | -0.00022  |

$$\therefore |x_{14} - x_{13}| = |1.324665431 - 1.324604238| = 6.1 \times 10^{-5} < 10^{-4}$$

So, 1.324665431 is the approximated solution.

**Ques - 03:**

$$2x - 3 \sin x - 5 = 0; \varepsilon = 10^{-3}$$

Soln: Let,  $f(x) = 2x - 3 \sin x - 5$

$$f(0) = 0 - 0 - 5 = -5 < 0$$

$$f(3) = 6 - 0.157007 - 5 = 0.5766 > 0$$

The function is continuous and the root lies in interval  $[0, 3]$

| Iteration | a        | b       | c = (a+b)/2 | f(c)     |
|-----------|----------|---------|-------------|----------|
| 1         | 0.00     | 3.00    | 1.5         | -1.992   |
| 2         | 1.5      | 3.00    | 2.25        | -2.8842  |
| 3         | 2.25     | 3.00    | 2.625       | -1.2317  |
| 4         | 2.625    | 3.00    | 2.8125      | -0.3445  |
| 5         | 2.8125   | 3.00    | 2.90625     | 0.11297  |
| 6         | 2.8125   | 2.90625 | 2.859375    | -0.1167  |
| 7         | 2.859375 | 2.90625 | 2.8828125   | -0.00207 |

| Iteration | a           | b           | c = (a+b)/2 | f(c)       |
|-----------|-------------|-------------|-------------|------------|
| 8         | 2.8828125   | 2.90625     | 2.89453125  | 0.0553     |
| 9         | 2.8828125   | 2.89453125  | 2.888671875 | 0.0266645  |
| 10        | 2.8828125   | 2.888671875 | 2.885742188 | -0.0020123 |
| 11        | 2.8828125   | 2.885742188 | 2.884277344 | -0.0197    |
| 12        | 2.884277344 | 2.885742188 | 2.885009766 | 0.008689   |

$$\therefore |x_{12} - x_{11}| = |2.885009766 - 2.884277344| = 7.32 \times 10^{-4} < 10^{-3}$$

$\therefore$  The approximate root solution is 2.8850097.

#### **Ques - 4:**

$$x^3 - 4x - 5 = 0; \varepsilon = 10^{-4}$$

Soln: Given that,  $x^3 - 4x - 5 = 0$

$$\text{let, } f(x) = x^3 - 4x - 5$$

$$f(0) = 0 - 0 - 5 = -5 < 0$$

$$f(3) = 27 - 12 - 5 = 10 > 0$$

The function is continuous and the root lies in interval  $[0, 3]$ .

The iterations for the functions are:  $[\varepsilon = 10^{-4}]$

| Iteration | a           | b           | c = (a+b)/2 | f(c)        |
|-----------|-------------|-------------|-------------|-------------|
| 1         | 0.00        | 3.00        | 1.5         | -7.625      |
| 2         | 1.5         | 3.00        | 2.25        | -2.6093     |
| 3         | 2.25        | 3.00        | 2.625       | 2.5878      |
| 4         | 2.25        | 2.625       | 2.4375      | -0.2678     |
| 5         | 2.4375      | 2.625       | 2.53125     | 1.0932      |
| 6         | 2.4375      | 2.53125     | 2.484375    | 0.3963      |
| 7         | 2.4375      | 2.484375    | 2.4609375   | 0.0602      |
| 8         | 2.4375      | 2.4609375   | 2.44921875  | -0.1098     |
| 9         | 2.44921875  | 2.4609375   | 2.455078125 | -0.0225     |
| 10        | 2.455078125 | 2.4609375   | 2.458007813 | 0.01876     |
| 11        | 2.455078125 | 2.458007813 | 2.456542969 | -0.001909   |
| 12        | 2.456542969 | 2.458007813 | 2.457275391 | 0.008424    |
| 13        | 2.456542969 | 2.457275391 | 2.45690918  | 0.003256    |
| 14        | 2.456542969 | 2.45690918  | 2.456726075 | 0.0006733   |
| 15        | 2.456542969 | 2.456726075 | 2.456634386 | -0.00062009 |

$$\therefore |x_{15} - x_{14}| = |2.456634386 - 2.456726075| = 9.1 \times 10^{-5} < 10^{-4}$$

$\therefore$  2.456634 is the approximate solution.

#### **Ques - 05:**

$$x \cdot e^x - 1 = 0; \varepsilon = 10^{-2}$$

Solution: Let,  $f(x) = x \cdot e^x - 1$

$$f(0) = 0 - 1 = -1 < 0$$

$$f(1) = 2.71828 - 1 = 1.71828$$

The function is continuous and the root lies in interval  $[0, 1]$ .

The iteration for the functions are:  $\epsilon = 10^{-2}$

| Iteration | a      | b      | $c = (a+b)/2$ | f(c)     |
|-----------|--------|--------|---------------|----------|
| 1         | 0.00   | 1.00   | 0.5           | -0.1756  |
| 2         | 0.5    | 1.00   | 0.75          | 0.5877   |
| 3         | 0.5    | 0.75   | 0.625         | 0.1676   |
| 4         | 0.5    | 0.625  | 0.5625        | -0.01278 |
| 5         | 0.5625 | 0.625  | 0.5938        | 0.0752   |
| 6         | 0.5625 | 0.5938 | 0.5782        | 0.0308   |
| 7         | 0.5625 | 0.5782 | 0.5704        | 0.00902  |

$$\therefore |x_7 - x_6| = |0.5704 - 0.5782| = 7.8 \times 10^{-3} < 10^{-2}$$

$\therefore 0.5704$  is the approximate solution.

## False Position Method / Regula Falsi Method

### Algorithm of False Position Method:

- 1) Define a function  $f(x)$ .
- 2) Find two points  $a$  and  $b$  so that  $f(a) < 0$  and  $f(b) > 0$ .
- 3) Define a new point,  $c = [a \cdot f(b) - b \cdot f(a)] / [f(b) - f(a)]$ .
- 4) If  $f(c) < 0$ , then set  $a = c$ , otherwise set  $b = c$ .
- 5) Continuing the process till the desired place of decimals is not obtained.

### Question - 01:

Determine the root of  $2x^3 - 2x - 5 = 0$  for  $x \in [1.2, 1.8]$  using False Position Method.  $\epsilon = 10^{-6}$

Soln: Let,  $f(x) = 2x^3 - 2x - 5 = 0$

$$f(1.2) = 2 \cdot (1.2)^3 - 2 \cdot 1.2 - 5 = -3.944 < 0$$

$$f(1.8) = 2 \cdot (1.8)^3 - 2 \cdot 1.8 - 5 = 3.064 > 0$$

The given function is continuous and the root lies in the interval  $[1.2, 1.8]$

| Iteration | a           | b   | $c = [a \cdot f(b) - b \cdot f(a)] / [f(b) - f(a)]$ | f(c)          |
|-----------|-------------|-----|-----------------------------------------------------|---------------|
| 1         | 1.2         | 1.8 | 1.53767123                                          | -0.803901     |
| 2         | 1.53767123  | 1.8 | 1.592193401                                         | -0.111712058  |
| 3         | 1.592193401 | 1.8 | 1.59950342                                          | -0.014631989  |
| 4         | 1.59950342  | 1.8 | 1.600456332                                         | -0.001901409  |
| 5         | 1.600456332 | 1.8 | 1.600580085                                         | -0.00024683   |
| 6         | 1.600580085 | 1.8 | 1.600596152                                         | -0.000032001  |
| 7         | 1.600596152 | 1.8 | 1.600598235                                         | -0.0000041442 |

| Iteration | a           | b   | $c = [a \cdot f(b) - b \cdot f(a)] / [f(b) - f(a)]$ | f(c)           |
|-----------|-------------|-----|-----------------------------------------------------|----------------|
| 8         | 1.600598235 | 1.8 | 1.600598505                                         | -0.00000005374 |

$$\therefore |x_8 - x_7| = |1.600598505 - 1.600598235| = 2.7 \times 10^{-7} < 10^{-6}$$

The approximated solution is 1.600598505.

### **Question - 02:**

$$x^3 - 2x - 5 = 0; \epsilon = 10^{-4} \text{ [False Position]}$$

$$\text{Soln: let, } f(x) = x^3 - 2x - 5$$

$$f(0) = 0 - 0 - 5 = -5 < 0$$

$$f(2.1) = (2.1)^3 - (2 \times 2.1) - 5 = 0.061 > 0$$

The given function is continuous and the root lies in interval [0, 2.1]

| Iteration | a           | b   | c           | f(c)          |
|-----------|-------------|-----|-------------|---------------|
| 1         | 0.00        | 2.1 | 2.074688797 | -0.21922488   |
| 2         | 2.074688797 | 2.1 | 2.094490099 | -0.00068508   |
| 3         | 2.094490099 | 2.1 | 2.094551284 | -0.0000022098 |
| 4         | 2.094551284 | 2.1 | 2.094619773 | -0.0007622    |

$$\therefore |x_4 - x_3| = |2.094619773 - 2.094551284| = 6.8 \times 10^{-5} < 10^{-4}$$

The approximated solution is 2.094619.

### **Ques - 3:**

$$3x - \cos x - 1 = 0; \epsilon = 10^{-3} \text{ [False Position]}$$

$$\text{Soln: Let, } f(x) = 3x - \cos x - 1$$

$$f(0) = 0 - \cos 0 - 1 = -2 < 0$$

$$f(1) = (3 \times 1) - \cos 1 - 1 = 1.4596 > 0$$

The function is continuous and the root lies in interval [0, 1]

| Iteration | a           | b    | c           | f(c)         |
|-----------|-------------|------|-------------|--------------|
| 1         | 0.00        | 1.00 | 0.5781015   | -0.103197013 |
| 2         | 0.5781015   | 1.00 | 0.60596095  | -0.004072315 |
| 3         | 0.60596095  | 1.00 | 0.607057268 | -0.000158455 |
| 4         | 0.607057268 | 1.00 | 0.607099921 | -0.000006164 |

$$\therefore |x_4 - x_3| = |0.607099921 - 0.607057268| = 4.2 \times 10^{-5} < 10^{-3}$$

The approximated solution is 0.6071.

### **Ques - 4:**

$$x^2 - \ln x - 12 = 0; \epsilon = 10^{-6} \text{ [False Position]}$$

$$\text{Soln: Let, } f(x) = x^2 - \ln x - 12$$

$$f(1) = (1)^2 - \ln(1) - 12 = -11 < 0$$

$$f(4) = (4)^2 - \ln(4) - 12 = 2.6137056 > 0$$

The given function is continuous and the root lies in interval [1, 4].

| Iteration | a           | b    | c           | f(c)         |
|-----------|-------------|------|-------------|--------------|
| 1         | 1.00        | 4.00 | 3.424027667 | -1.506852    |
| 2         | 3.424027667 | 4.00 | 3.634655711 | -0.0797922   |
| 3         | 3.634655711 | 4.00 | 3.645478671 | -0.003972    |
| 4         | 3.645478671 | 4.00 | 3.646016613 | -0.00019809  |
| 5         | 3.646016613 | 4.00 | 3.646043439 | -0.000009831 |
| 6         | 3.646043439 | 4.00 | 3.64604477  | -0.000000488 |

$$\therefore |x_6 - x_5| = 3.8 \times 10^{-7} < 10^{-6}$$

3.64604477 is the approximated solution.

### **Question - 05:**

$x \cdot e^x = 1$ ,  $x \in [0, 1]$ , correct upto 4 decimal places.

Soln: Given that,  $x \cdot e^x = 1 \Rightarrow x \cdot e^x - 1 = 0$

Let,  $f(x) = x \cdot e^x - 1$

$$f(0) = 0 \cdot e^0 - 1 = 0 - 1 = -1 < 0$$

$$f(1) = 1 \cdot e^1 - 1 = 1.718280 > 0$$

Since,  $f(0) \times f(1) < 0$ .

The given function is continuous and the root lies in interval [0, 1].

The iterations for the functions are:

| Iteration | a           | b    | c           | f(c)         |
|-----------|-------------|------|-------------|--------------|
| 1         | 0.00        | 1.00 | 0.3678905   | -0.1685145   |
| 2         | 0.3678905   | 1.00 | 0.503318162 | -0.1674105   |
| 3         | 0.503318162 | 1.00 | 0.547415156 | -0.0536403   |
| 4         | 0.547415156 | 1.00 | 0.561116657 | -0.016570958 |
| 5         | 0.561116657 | 1.00 | 0.565308978 | -0.0120339   |
| 6         | 0.565308978 | 1.00 | 0.566585613 | -0.0129388   |
| 7         | 0.566585613 | 1.00 | 0.566973893 | -0.0004689   |
| 8         | 0.566973893 | 1.00 | 0.567091    | -0.0001424   |
| 9         | 0.567091    | 1.00 | 0.567126    | -0.0000452   |
| 10        | 0.567126    | 1.00 | 0.567137    | -0.0000162   |

$$\text{At the 10th iteration, } |x_{10} - x_9| = |0.567137 - 0.567126| = 1.1 \times 10^{-5} < 10^{-4}$$

The approximated solution is 0.5671, which is correct upto 4 decimal places.

### **Question - 06:**

$\cos x = 3x - 1$ ;  $x \in [0, 1]$ ; correct upto 5 decimal places.

Solution: Given that,  $\cos x = 3x - 1 \Rightarrow \cos x - 3x + 1 = 0$

Let,  $f(x) = \cos x - 3x + 1$

$$f(0) = \cos 0 - (3 \times 0) + 1 = 2 > 0$$

$$f(1) = \cos 1 - (3 \times 1) + 1 = -1.4596976 < 0$$

$$\text{Since, } f(0) \times f(1) < 0$$

The given function is continuous, and the root lies in interval  $[0, 1]$ .

The iterations for the given functions are:

| Iteration | a           | b    | c           | f(c)         |
|-----------|-------------|------|-------------|--------------|
| 1         | 0.00        | 1.00 | 0.57808519  | 0.103254906  |
| 2         | 0.57808519  | 1.00 | 0.605958573 | -0.0040807   |
| 3         | 0.605958573 | 1.00 | 0.607057144 | -0.0001589   |
| 4         | 0.607057144 | 1.00 | 0.607099914 | -0.00000618  |
| 5         | 0.607099914 | 1.00 | 0.607101583 | -0.000000231 |
| 6         | 0.607101583 | 1.00 | 0.607101645 | 0.0000000082 |

$$\text{At the 6th point, } |x_6 - x_5| = |0.607101645 - 0.607101583| = 6.2 \times 10^{-8} < 10^{-5}$$

Correct upto 5 decimal places, 0.60710 is the approximated solution.

#### **Question - 07:**

$$\cos x - x \cdot e^x = 0; x \in [0, 1], \text{ upto 4 decimal places.}$$

$$\text{Solution: Given that, } \cos x - x \cdot e^x = 0$$

$$\text{Let, } f(x) = \cos x - x \cdot e^x$$

$$f(0) = \cos 0 - 0 \cdot e^0 = 1 > 0$$

$$f(1) = \cos 1 - 1 \cdot e^1 = -2.177979 < 0$$

$$\text{Since, } f(0) \times f(1) < 0$$

The interval is valid and the function is continuous.

The iterations for the functions are:

| Iteration | a          | b    | c          | f(c)        |
|-----------|------------|------|------------|-------------|
| 1         | 0.00       | 1.00 | 0.3146652  | 0.519871964 |
| 2         | 0.3146652  | 1.00 | 0.446728   | 0.203544    |
| 3         | 0.446728   | 1.00 | 0.494015   | 0.070802    |
| 4         | 0.494015   | 1.00 | 0.509945   | 0.023608    |
| 5         | 0.509945   | 1.00 | 0.51520027 | 0.007762    |
| 6         | 0.51520027 | 1.00 | 0.516922   | 0.002539    |
| 7         | 0.516922   | 1.00 | 0.517484   | 0.000829    |
| 8         | 0.517484   | 1.00 | 0.517667   | 0.000272    |
| 9         | 0.517667   | 1.00 | 0.517728   | 0.0000909   |
| 10        | 0.517728   | 1.00 | 0.517746   | 0.0000344   |

$$\text{At the 10th point, } |x_{10} - x_9| = |0.517746 - 0.517728| = 1.8 \times 10^{-5} < 10^{-4}$$

Correct upto 4 decimal places, 0.5177 is the approximated solution.

#### **Ques - 08:**

$$e^x - 3x = 0; x \in [1.50, 1.52]$$

Solution: Given that,  $e^x - 3x = 0$ ;  $x \in [1.50, 1.52]$

$$f(1.50) = e^{1.50} - (3 \times 1.50) = -0.01831093 < 0$$

$$f(1.52) = e^{1.52} - (3 \times 1.52) = 0.01225195 > 0$$

Since,  $f(1.50) \times f(1.52) < 0$

The given function is continuous and the interval is valid.

The iterations for the given functions:

| Iteration | a           | b    | c           | f(c)         |
|-----------|-------------|------|-------------|--------------|
| 1         | 1.50        | 1.52 | 1.512004    | -0.000199426 |
| 2         | 1.512004    | 1.52 | 1.511992964 | -0.000217493 |
| 3         | 1.511992964 | 1.52 | 1.512132922 | -0.000002509 |
| 4         | 1.512132922 | 1.52 | 1.512134533 | -0.000000029 |

$$\therefore |x_4 - x_3| = |1.512134533 - 1.512132922| = 0.000001611 < 0.00005$$

1.5121 is the approximated solution.

## Newton-Raphson Method

### Algorithm of Newton-Raphson Method:

- 1) Define a function  $f(x)$ .
- 2) Find points a and b such that  $a < b$  and  $f(a) \times f(b) < 0$ .
- 3) Take the interval  $[a, b]$  and find next value,  $x_0 = (a + b) / 2$ .
- 4) Find the derivative of the function  $f'(x)$ .
- 5) Find  $x_{n+1} = x_n - f(x_n) / f'(x_n)$ .
- 6) If  $f(x_n) = 0$  then  $x_n$  is an exact root, otherwise  $x_n = x_{n+1}$ .
- 7) Repeat the step 5 and 6 until  $f(x_i) = 0$ ,  $i = 1, 2, 3 \dots$  or  $|f(x_i)| \leq \text{Accuracy}$ .

### Question - 01:

Find the positive root of  $x = \cos x$  for  $x \in [0, 1]$  using Newton-Raphson method.

Solution: Let,  $f(x) = x - \cos x$

$$\therefore f'(x) = 1 + \sin x$$

$$\text{Here, } f(0) = 0 - \cos 0 = -1 < 0$$

$$f(1) = 1 - \cos 1 = 0.45969 > 0$$

The given function is continuous, and the root lies in the interval  $[0, 1]$ .

$$\therefore x_0 = (0 + 1) / 2 = 0.5$$

By Newton-Raphson method, we have,

$$x_{n+1} = x_n - f(x_n) / f'(x_n)$$

$$\Rightarrow x_{n+1} = x_n - (x_n - \cos x_n) / (1 + \sin x_n)$$

$$= (x_n + x_n \cdot \sin x_n - x_n + \cos x_n) / (1 + \sin x_n)$$

$$= (x_n \cdot \sin x_n + \cos x_n) / (1 + \sin x_n)$$

Taking,  $x_0 = 0.5$ , we get,

$$\text{when } n = 0; x_1 = (x_0 \sin x_0 + \cos x_0) / (1 + \sin x_0) = (0.5 \sin 0.5 + \cos 0.5) / (1 + \sin 0.5)$$

$$x_1 = 0.755222417$$

$$n = 1; x_2 = (x_1 \sin x_1 + \cos x_1) / (1 + \sin x_1) = 0.739014166$$

$$n = 2; x_3 = (x_2 \sin x_2 + \cos x_2) / (1 + \sin x_2) = 0.739085133$$

$$n = 3; x_4 = (x_3 \sin x_3 + \cos x_3) / (1 + \sin x_3) = 0.739085133$$

$$\therefore \text{Since, } |x_4 - x_3| = 4 \times 10^{-9} < 10^{-8}.$$

Hence, the required root is 0.739085133.

## Homework

**1)  $x^3 - 2x^2 - 4 = 0$ ,  $x \in [2, 3]$ , correct upto 5 decimal places.**

Solution: Given that,  $x^3 - 2x^2 - 4 = 0$

$$\text{Let, } f(x) = x^3 - 2x^2 - 4$$

$$\therefore f'(x) = 3x^2 - 4x$$

$$\text{Here, } f(2) = (2)^3 - 2 \cdot (2)^2 - 4 = -4 < 0$$

$$f(3) = (3)^3 - 2 \cdot (3)^2 - 4 = 5 > 0$$

$$\therefore \text{Since, } f(2) \times f(3) < 0.$$

The given function is continuous, and the root lies in the interval  $[2, 3]$

$$\therefore x_0 = (2 + 3) / 2 = 5/2 = 2.5$$

By Newton-Raphson method, we have,

$$x_{n+1} = x_n - f(x_n) / f'(x_n)$$

$$\Rightarrow x_{n+1} = x_n - (x_n^3 - 2x_n^2 - 4) / (3x_n^2 - 4x_n)$$

$$= (3x_n^3 - 4x_n^2 - x_n^3 + 2x_n^2 + 4) / (3x_n^2 - 4x_n)$$

$$= (2x_n^3 - 2x_n^2 + 4) / (3x_n^2 - 4x_n)$$

Taking,  $x_0 = 2.5$ , we get,

$$\text{when } n = 0; x_1 = (2x_0^3 - 2x_0^2 + 4) / (3x_0^2 - 4x_0) = [2 \times (2.5)^3 - 2 \times (2.5)^2 + 4] / [3 \times (2.5)^2 - 4 \times 2.5]$$

$$\therefore x_1 = 2.6$$

$$n = 1; x_2 = (2x_1^3 - 2x_1^2 + 4) / (3x_1^2 - 4x_1) = 2.594331984$$

$$n = 2; x_3 = (2x_2^3 - 2x_2^2 + 4) / (3x_2^2 - 4x_2) = 2.594313017$$

$$n = 3; x_4 = (2x_3^3 - 2x_3^2 + 4) / (3x_3^2 - 4x_3) = 2.594313016$$

$$\text{Since, } |x_4 - x_3| = 10^{-9} < 10^{-5}.$$

Hence the required root is 2.59431 [correct up to 5 decimal places].

**2)  $x^4 + x^2 - 80 = 0$ ;  $x \in [2, 3]$ , correct upto 3 decimal places in this actionS**

$\therefore$  3 decimal places: 2.908.

Hence, the required root is 2.908 [correct up to 3 decimal places].

**Question – 03:**

$e^{-x} = 10x$ ,  $x \in [0, 0.2]$ , correct up to 8 decimal places.

Solution: Given that,  $e^{-x} = 10x \Rightarrow e^{-x} - 10x = 0$

Let,  $f(x) = e^{-x} - 10x$

$\therefore f'(x) = -e^{-x} - 10$

Here,  $f(0) = e^0 - 10 \times 0 = 1 > 0$

$f(0.2) = e^{-0.2} - (10 \times 0.2) = -1.18126 < 0$

Since,  $f(0) \times f(0.2) < 0$

The given function is continuous and the root lies in the interval  $[0, 0.2]$ .

$\therefore x_0 = (0 + 0.2)/2 = 0.1$

By the Newton-Raphson method, we know,

$$x_{n+1} = x_n - f(x_n) / f'(x_n)$$

$$= x_n - (e^{-x_n} - 10x_n) / (-e^{-x_n} - 10)$$

$$= (e^{-x_n} \cdot x_n + e^{-x_n}) / (e^{-x_n} + 10)$$

Taking,  $x_0 = 0.1$

When  $n = 0$ , then,

$$x_1 = (e^{-x_0} \cdot x_0 + e^{-x_0}) / (e^{-x_0} + 10) = 0.09127336$$

$$n = 1 : x_2 = (e^{-x_1} \cdot x_1 + e^{-x_1}) / (e^{-x_1} + 10) = 0.091276527$$

$$n = 2 : x_3 = (e^{-x_2} \cdot x_2 + e^{-x_2}) / (e^{-x_2} + 10) = 0.091276527$$

Correct upto 8 decimal places:  $x = 0.09127652$

The root is  $x = 0.09127652$ .

---

## Exercise

### Q – 01:

$$2x^3 - 3x - 6 = 0 ; [1, 2]$$

Solution: Let,  $f(x) = 2x^3 - 3x - 6$

$$f'(x) = 6x^2 - 3$$

$$\text{Here, } f(1) = 2 \times (1)^3 - (3 \times 1) - 6 = -7$$

$$f(2) = 2 \times (2)^3 - (3 \times 2) - 6 = 4 > 0$$

Since,  $f(1) \times f(2) < 0$

The given function is continuous and the root lies in the interval  $[1, 2]$ .

$$\therefore x_0 = (1+2)/2 = 1.5$$

By the Newton-Raphson method, we know,

$$x_{n+1} = x_n - f(x_n) / f'(x_n)$$

$$= x_n - (2x_n^3 - 3x_n - 6) / (6x_n^2 - 3)$$

$$= (6x_n^3 - 3x_n - 2x_n^3 + 3x_n + 6) / (6x_n^2 - 3)$$

$$\therefore x_{n+1} = (4x_n^3 + 6) / (6x_n^2 - 3)$$

Taking,  $x_0 = 1.5$

$$n = 0 : x_1 = (4x_0^3 + 6) / (6x_0^2 - 3) = (4 \times (1.5)^3 + 6) / (6 \times (1.5)^2 - 3) = 1.857142857$$

$$n = 1 : x_2 = (4x_1^3 + 6) / (6x_1^2 - 3) = 1.787114846$$

$$n = 2 : x_3 = (4x_2^3 + 6) / (6x_2^2 - 3) = 1.783776483$$

$$n = 3 : x_4 = (4x_3^3 + 6) / (6x_3^2 - 3) = 1.783769061$$

Correct upto 5 decimal places, the root is 1.78376.

---

### Q – 02:

$$x \cdot e^x - \cos x = 0 ; [0, 1]$$

Solution: Let,  $f(x) = x \cdot e^x - \cos x$

$$f'(x) = x \cdot e^x + e^x + \sin x$$

$$\text{Here, } f(0) = 0 \times e^0 - \cos 0 = -1 < 0$$

$$f(1) = 1 \times e^1 - \cos 1 = 0.58257211 > 0$$

Since,  $f(0) \times f(1) < 0$

The given function is continuous and the root lies in the interval  $[0, 1]$ .

$$\therefore x_0 = (0+1)/2 = 0.5$$

By Newton-Raphson method, we have,

$$x_{n+1} = x_n - f(x_n) / f'(x_n)$$

$$\begin{aligned}
&= x_n - (x_n \cdot e^{x_n} - \cos x_n) / (e^{x_n} + x_n \cdot e^{x_n} + \sin x_n) \\
&= (e^{x_n} \cdot x_n + x_n^2 e^{x_n} + x_n \cdot \sin x_n - x_n \cdot e^{x_n} + \cos x_n) / (e^{x_n} + x_n \cdot e^{x_n} + \sin x_n) \\
&\therefore x_{n+1} = (x_n^2 \cdot e^{x_n} + x_n \cdot \sin x_n + \cos x_n) / (e^{x_n} + x_n \cdot e^{x_n} + \sin x_n)
\end{aligned}$$

Taking,  $x_0 = 0.5$

$$n = 0 : x_1 = (x_0^2 \cdot e^{x_0} + x_0 \cdot \sin x_0 + \cos x_0) / (e^{x_0} + x_0 \cdot e^{x_0} + \sin x_0) = 0.518026009$$

$$n = 1 : x_2 = 0.517757424$$

$$n = 2 : x_3 = 0.517757363$$

$$n = 3 : x_4 = 0.517757363$$

Correct upto 5 decimal places, 0.51775 is the root.

### Q – 03:

$$3x - \cos x - 1 = 0 ; [0, 1]$$

Solution: Let,  $f(x) = 3x - \cos x - 1$

$$f'(x) = 3 + \sin x$$

Here,

$$f(0) = 3 \times 0 - \cos 0 - 1 = -2 < 0$$

$$f(1) = 3 \times 1 - \cos 1 - 1 = 1.459697 > 0$$

$$\text{Since, } f(0) \times f(1) < 0$$

The given function is continuous and the root lies in the interval  $[0, 1]$ .

$$\therefore x_0 = (0+1)/2 = 0.5$$

By Newton-Raphson method, we have,

$$\begin{aligned}
x_{n+1} &= x_n - f(x_n) / f'(x_n) \\
&= x_n - (3x_n - \cos x_n - 1) / (3 + \sin x_n) \\
&= (3x_n + x_n \cdot \sin x_n - 3x_n + \cos x_n + 1) / (3 + \sin x_n) \\
&\therefore x_{n+1} = (x_n \cdot \sin x_n + \cos x_n + 1) / (3 + \sin x_n)
\end{aligned}$$

Taking,  $x_0 = 0.5$

$$n = 0 : x_1 = (x_0 \cdot \sin x_0 + \cos x_0 + 1) / (3 + \sin x_0) = (0.5 \times \sin 0.5 + \cos 0.5 + 1) / (3 + \sin 0.5)$$

$$\therefore x_1 = 0.608518649$$

$$n = 1 : x_2 = 0.607101878$$

$$n = 2 : x_3 = 0.607101648$$

$$n = 3 : x_4 = 0.607101648$$

Correct upto 5 decimal places, 0.60710 is the root.

---

**Q – 04:**

$$10x - \sin x - 6 = 0 ; [0, 1]$$

Solution: Let,  $f(x) = 10x - \sin x - 6$

$$f'(x) = 10 - \cos x$$

$$f(0) = (10 \times 0) - \sin 0 - 6 = -6 < 0$$

$$f(1) = (10 \times 1) - \sin 1 - 6 = 3.15852 > 0$$

$$\text{Since, } f(0) \times f(1) < 0$$

The given function is continuous and the root lies in the interval  $[0, 1]$ .

$$\therefore x_0 = (0+1)/2 = 0.5$$

By Newton-Raphson Method, we have,

$$x_{n+1} = x_n - f(x_n) / f'(x_n)$$

$$= x_n - (10x_n - \sin x_n - 6) / (10 - \cos x_n)$$

$$= (10x_n - x_n \cdot \cos x_n - 10x_n + \sin x_n + 6) / (10 - \cos x_n)$$

$$\therefore x_{n+1} = (\sin x_n - x_n \cdot \cos x_n + 6) / (10 - \cos x_n)$$

Taking,  $x_0 = 0.5$

$$n = 0 : x_1 = (\sin x_0 - x_0 \cdot \cos x_0 + 6) / (10 - \cos x_0) = (\sin 0.5 - 0.5 \cdot \cos 0.5 + 6) / (10 - \cos 0.5)$$

$$\therefore x_1 = 0.662174724$$

$$n = 1 : x_2 = 0.661424146$$

$$n = 2 : x_3 = 0.661424128$$

$$n = 3 : x_4 = 0.661424128$$

Correct upto 6 decimal places, 0.661424 is the root.

# Romberg Integration

Use Romberg integration to compute  $R_{3,3}$  for the following integral:

$$\int_0^1 x^2 \cdot e^{-x} dx$$

**Romberg table structure:**

| k | R <sub>k,1</sub> | R <sub>k,2</sub> | R <sub>k,3</sub> |
|---|------------------|------------------|------------------|
| 1 | R <sub>1,1</sub> |                  |                  |
| 2 | R <sub>2,1</sub> | R <sub>2,2</sub> |                  |
| 3 | R <sub>3,1</sub> | R <sub>3,2</sub> | R <sub>3,3</sub> |

---

**Solution:**

Here,  $f(x) = x^2 \cdot e^{-x}$ ,  $a = 0$  and  $b = 1$ .

and,  $h_k = (b - a) / 2^{k-1}$

For,  $k = 1$ ,  $h_1 = (1 - 0) / 2^{1-1} = 1/1 = 1$

For,  $k = 2$ ,  $h_2 = (1 - 0) / 2^{2-1} = 1/2 = 0.5$

For,  $k = 3$ ,  $h_3 = (1 - 0) / 2^{3-1} = 1/4 = 0.25$

$$\begin{aligned} R_{1,1} &= (h_1/2)[f(a) + f(b)] \\ &= (1/2)[f(0) + f(1)] = \frac{1}{2}[0 + 0.367879] = 0.1839397206 \end{aligned}$$

$$\begin{aligned} R_{2,1} &= (h_2/2)[f(a) + f(b) + 2f(a + h_2)] \\ &= (0.5/2)[f(0) + f(1) + 2f(0.5)] = 0.1677861928 \end{aligned}$$

$$\begin{aligned} R_{3,1} &= (h_3/2)[f(a) + f(b) + 2\{f(a + h_3) + f(a + 2h_3) + f(a + 3h_3)\}] \\ &= (0.25/2)[f(0) + f(1) + 2\{f(0.25) + f(0.5) + f(0.75)\}] \\ &= 0.1624884057 \end{aligned}$$

---

**Improving the composite trapezoidal rule (CTR) using Richardson extrapolation:**

$$\begin{aligned} R_{2,2} &= R_{2,1} + (R_{2,1} - R_{1,1}) / (4^{2-1} - 1) \\ &= 0.16240168 \end{aligned}$$

$$R_{3,2} = R_{3,1} + (R_{3,1} - R_{2,1}) / (4^{2-1} - 1)$$

$$= 0.1607224759$$

$$R_{3,3} = R_{3,2} + (R_{3,2} - R_{2,2}) / (4^{3-1} - 1)$$

$$= 0.1606105289$$

---


$$\therefore R_{3,3} \approx \mathbf{0.16061053}$$